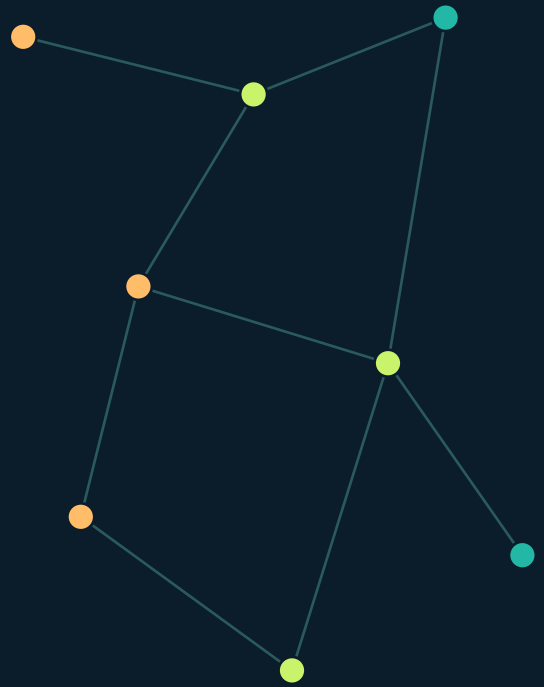
IS 507 • TEAM PROJECT

To predict the harvest, learn the data signals.

District-level crop-yield prediction in India: why it matters, what the system signals tell us, and what we can’t explain (yet!)

Aadi, Bagus and Shashank



Causal and Predictive Modeling of Indian Crop Yields and Agricultural Prices

Agricultural uncertainty becomes economic uncertainty

Forecasts matter because decisions are made before the final harvest is known.

~1 in 5

of national income comes from agriculture and allied activities

46.1%

of India's workforce depends on agriculture and allied activities

$382B

food-processing market, FY2024–25 estimate

A forecast is useful when it changes a decision, not only when it scores well.

HARVEST SIGNAL

How much is likely to be produced?

→

OPERATING DECISIONS

Procurement, production, inventory

→

SYSTEM-WIDE EXPOSURE

Food security, supply chains, price support

GOVERNMENT OF INDIA · ECONOMIC SURVEY 2025–26

Rice and wheat procurement has at times exceeded food-security needs, **leading to costly buffer stocks.**

Sources: Government of India, Economic Survey 2025–26, Ch. 6 (Box VI.5); IBEF, Food Processing Industry (FY25).

Our baseline: research shows “weather” improves forecasts

Our aim: predict crop yield and production from agro-meteorological and agro-economic variables.

PUBLISHED STUDY · ONE CROP, ONE DISTRICT

Dharmaraja et al. (2020)

441 → 51 kg/ha forecast error

Adding weather to a time-series model (ARIMAX) cut error by ~89%. Pearl millet, Alwar district; trained 1997–2013, tested 2014–16.

PUBLISHED STUDY · NATIONAL, A FEW CROPS, SMALL EVIDENCE

Birthal et al. (2014), Imbens et al. (2015)

Heat ↓ yield

Higher maximum temperatures reduce yields; more rain helps but does not fully offset the heat. Irrigation helps reduce the damage.

THE GAP

One crop in one district, or a few crops nationally. **Does it hold for yield and production of 23 crops across 602 districts?**

Sources: Dharmaraja et al. (2020), Agricultural Research 9:132–138; Birthal et al. (2014), Agricultural Economics Research Review 27(2):145–155.

One hypothesis, five research questions

HYPOTHESIS (SET BEFORE LOOKING AT AVAILABLE DATA)

H0 Agro-meteorological and agro-economic variables do not reduce forecast error vs. history alone.

H1 They lower MAE and/or RMSE on the same held-out years.

RQ1	How accurately can yield be predicted from historical yield alone, and how much do agro-meteorological and other controllable variables like irrigation and agri inputs improve out-of-sample accuracy?
RQ2	Do gains persist across crops, regions, and future years, or are they concentrated in specific crop-district systems?
RQ3	Do socioeconomic and infrastructure variables add predictive value beyond biophysical data, and can any relationship be interpreted causally?
RQ4	Is available price data sufficiently complete and aligned for valid crop-market-time forecasting?
RQ5	To what extent do central and state agricultural policy interventions correlate with improvements in district-level yield resilience and price stability, and how do projected structural gains from post-2017 policy scaling compare against historical baselines established through the data?

Research questions as stated in our project proposal. MAE = mean absolute error; RMSE = root mean squared error.

Crops × districts × yearly trends, many ways to model it

ICRISAT District-Level Database: raw files merged, cleaned and harmonised by us.

~27	~30	602	23	175 + 174
raw source files	years (1990–2019)	districts in 20 states	crops tracked	selected parameters

How we modelled it

1. **Covariance (R^2)** every variable × every crop × every district
2. **Two-way fixed effects** district and year held fixed
3. **Forecast** history only vs. history + environment

Ridge · ARIMA · Prophet · Decision tree (ID3) · Random forest · XGBoost

What made it hard

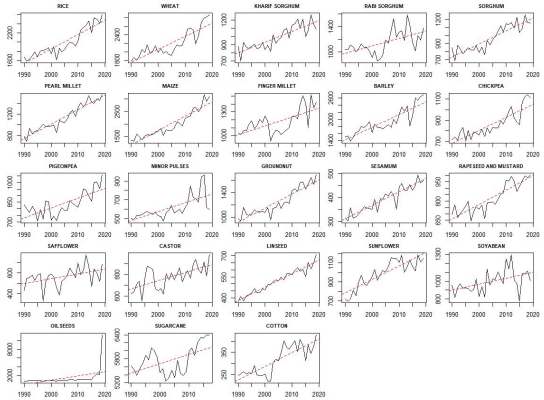
17–31%	yield values missing by crop; uneven missing codes
Mixed	granularity: monthly weather, yearly crops, census-year socio data
602 vs 535	district counts differ; boundaries redrawn
Uneven crop cycle	one crop season crosses two calendar years; prices are yearly

Source: ICRISAT District-Level Database for Indian Agriculture and Allied Sectors (via Tata-Cornell Institute). Imputation fitted on training years only.

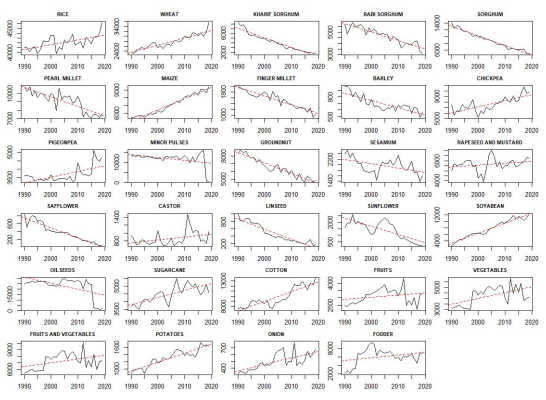
Annual yield went up for nearly every crop; area did not

All-India trends per crop, 1990–2019. Red dashed line = linear trend.

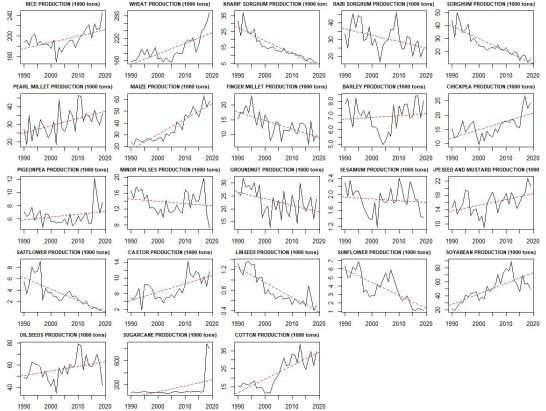
Yield (kg/ha)



Area (1,000 ha)



Production (1,000 t)



Read left to right: yield rises almost everywhere; area falls for coarse grains and minor oilseeds; production follows area.

Inputs matter differently by crop and by district

Example district: Darjeeling, West Bengal. Every district has one heatmap (see appendix).

R^2 of crop yield vs. each variable, 1990–2019



Strongest: Groundnut × fertilizer/ha (0.76) · Groundnut × nitrogen (0.72) · Soybean × potash (0.66)

Direction of change, 1990–2019

Crop	Production	Yield	Area	Comment
Rice	UP	UP	UP	
Wheat	UP	UP	UP	
Maize	UP	UP	UP	
Chickpea	UP	UP	UP	
Pigeonpea	UP	UP	UP	
Soybean	UP	UP	UP	
Sugarcane	UP	UP	UP	
Cotton	UP	UP	UP	
Pearl millet	UP	UP	DOWN	Area decreasing but yield is great its help production
Rapeseed and mustard	UP	UP	NO CLEAR	
Castor	UP	UP	NO CLEAR	
Kharif sorghum	DOWN	UP	DOWN	Area decreasing more than yield makes production decreasing
Sorghum	DOWN	UP	DOWN	
Finger millet	DOWN	UP	DOWN	
Safflower	DOWN	UP	DOWN	
Linseed	DOWN	UP	DOWN	
Sunflower	DOWN	UP	DOWN	

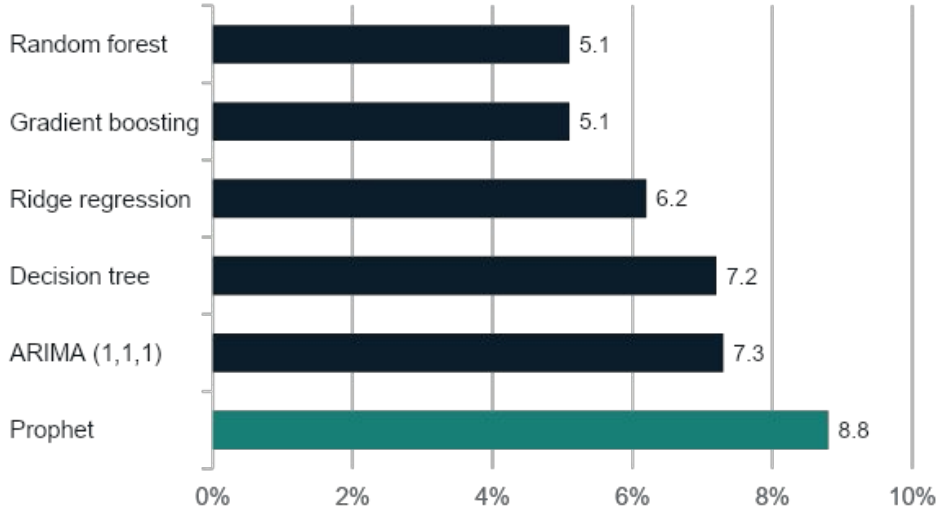
R^2 from simple regressions of district yield on each variable; association only. Table: team summary of the trend plots on the previous slide.

PRELIMINARY

Agro-meteorological attributes improve performance, but with caveats

Baseline: each model on history only. Then the same model + agro-meteorological variables. Test years 2018–19.

Lower pooled MAE than the history-only baseline (production)



6 of 6

models: lower MAE (by 5.1–8.8%)

2 of 6

models: lower MSE; large misses remain

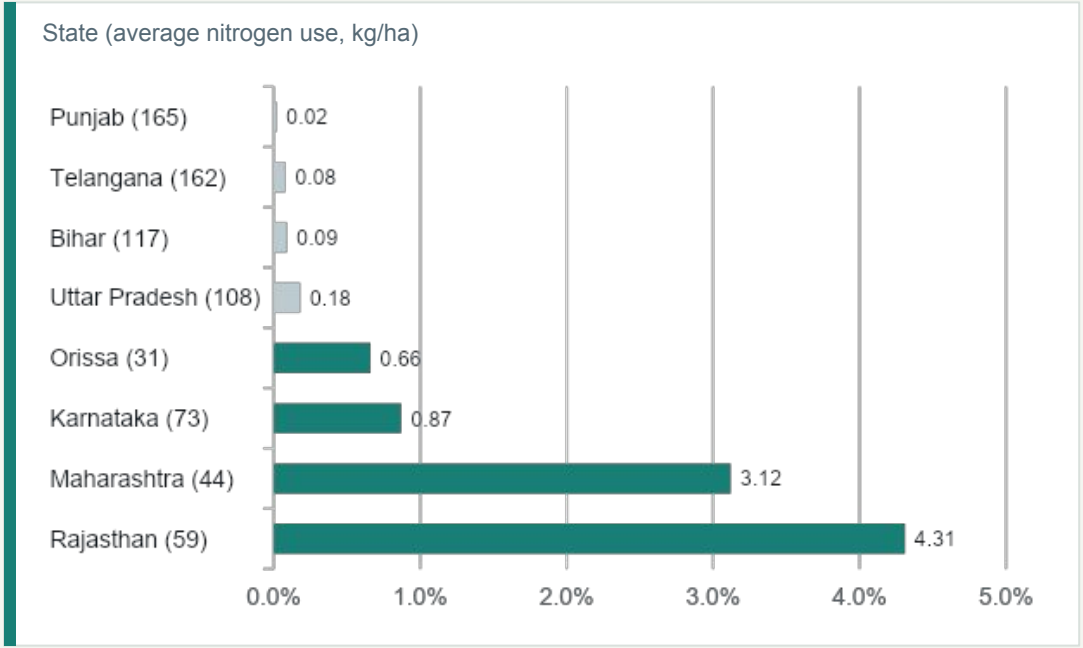
CAVEATS

National totals, 2 test years. Yield (kg/ha) is mixed: Prophet −24.6% MAE, ARIMA +45.9%.

Exploratory national crop-year runs, 23 crops, 2018–19 holdout. Use as the post-EDA experimental baseline, not a final validation claim.

Law of Diminishing Returns: Nitrogen is linked to higher yield only where farmers use little

Rice: change in yield per +10 kg/ha nitrogen, by state (district and year fixed effects).



+4.3%

Rajasthan rice, low nitrogen use (≈59 kg/ha)

≈ 0%

Punjab rice, high nitrogen use (≈165 kg/ha)

SAME FOR WHEAT

Madhya Pradesh +2.4% vs. Punjab ≈ 0%. Fits diminishing returns.

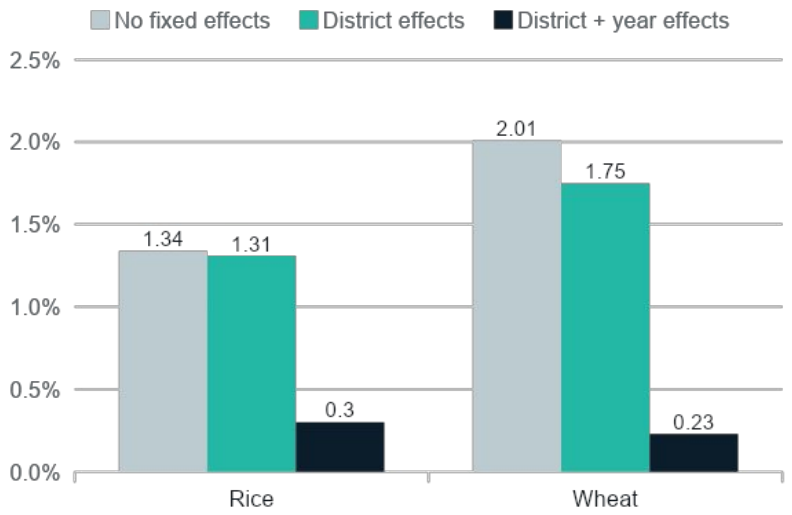
Our estimates: log yield on nitrogen per ha of gross cropped area and irrigated share, district + year fixed effects, errors clustered by district. Association only.

PRELIMINARY

Separating Signal from Noise : Most of the fertilizer "effect" is a shared national trend

District effects remove fixed differences between places; year effects remove shocks shared by all districts.

Yield change per +10 kg/ha nitrogen (year-on-year)



What survives fixed effects

+3.5%

yield per +10 pts irrigated share (rice, wheat)

−3.7%

rice yield per +1°C monsoon max temp

WHAT IT COULD MEAN FOR POLICY

Irrigation and heat resilience may matter more than extra fertilizer.

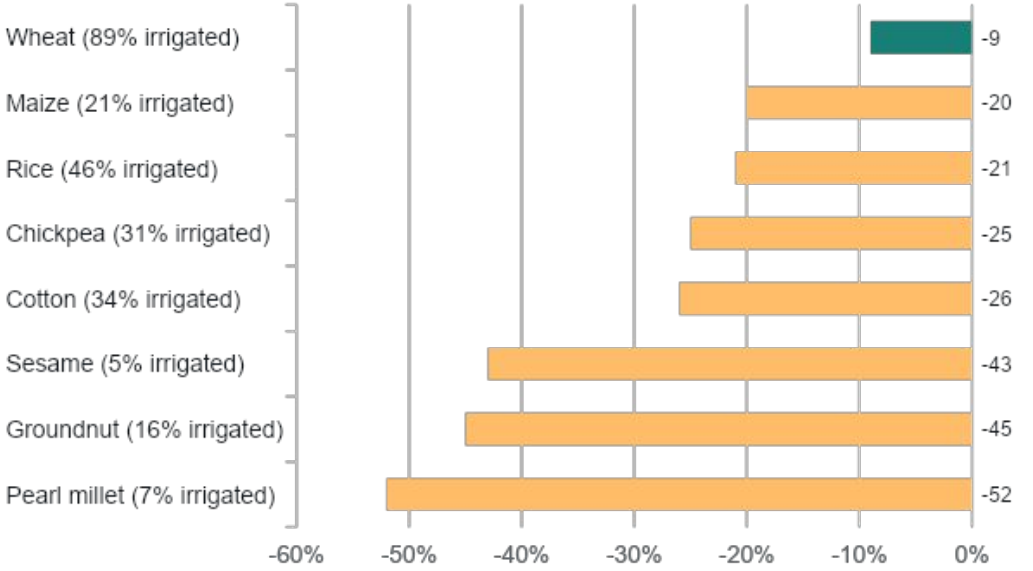
Target inputs to low-use states, not high-use states.

Test each policy directly; do not assume.

Our estimates, 1990–2019: log yield on N/ha, irrigated share, seasonal rain, max temp; SE clustered by district. Rice n = 10,844; wheat n = 8,761.

Irrigation buffered the 2002 drought; rainfed crops fell

Infrastructure mediates drought: production in 2002 vs. the 2001–03 average, by share of crop irrigated.



−52%

pearl millet production (7% irrigated)

−9%

wheat production (89% irrigated)

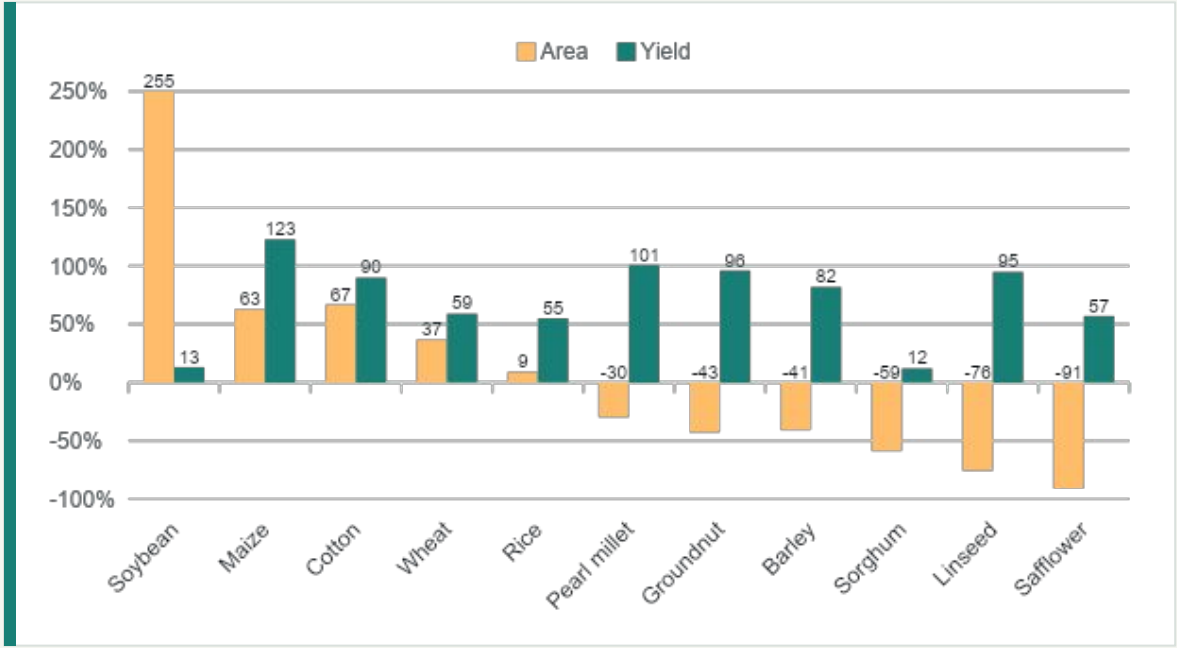
ALSO: 2000–2005

Area starts falling: linseed −45%, safflower −27%, barley −18%, sorghum −14%.

Our calculation. Irrigated share = irrigated ÷ crop area, 2001–03. Area change: 1997–99 vs 2003–05. 2002 had the lowest June–Sept rain (median ≈620 mm).

Yields rose for most crops; cultivated land decided production

Yield = tonnes per hectare; production = yield × cultivated area. Change 1990–92 vs. 2017–19 (all India).



27.8 → 14.2M ha

coarse grains area (sorghum, millets, barley): halved

15.6 → 35.1M ha

maize + soybean + cotton area: more than doubled

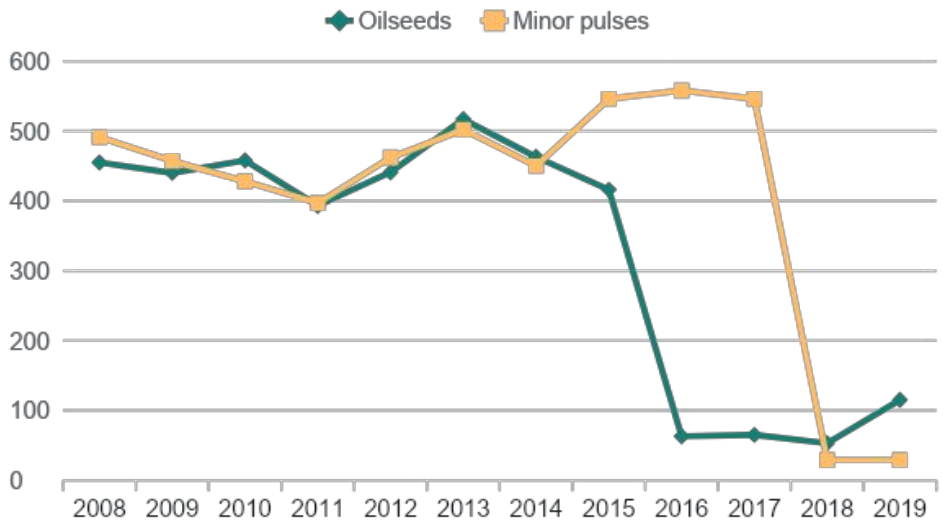
A shift from food grains to commercial crops. Descriptive, not causal.

Our calculation: national sums of district area and production; yield = production ÷ area. Sugarcane, oilseeds and minor pulses excluded (reporting breaks).

A few dramatic outliers and reporting breaks!

Before modelling, we checked how many districts report each crop each year.

Districts reporting the crop, per year



416 → 63

oilseeds districts reporting, 2015 → 2016

546 → 29

minor pulses districts reporting, 2017 → 2018

Sugarcane "+1,423%"

Yield jumps 10× in 2018 (7,944 → 79,318 kg/ha): a unit change. Real growth ≈ +40%.

Our calculation from ICRISAT district file; national sums of district values. Real sugarcane growth compares 1990–92 with 2015–17 averages.

Adding policy is the hardest step; here is our plan

Policies are written as text. To test them, we must turn them into data.

01
Collect
Central and state policy documents

→

02
Extract
Language model + human check

→

03
Link
District × year × crop table × optimization feature

→

04
Test
Difference-in-differences or fixed effects

Five fields per policy

Policy type	Timing	Target group	Geography	Implementation

WHY IT IS HARD

Announced is not the same as implemented.

Many policies overlap in the same years.

Exposure differs by district and crop.

SOCIO- AND AGRO-ECONOMIC LAYER

Population, literacy, farm workers, banks, markets, roads, farm size, livestock.

Mostly census years only: context, not yearly change.

Policy extraction follows the project proposal (Track C, H4). Socio-economic fields come from our 174-column merged table.

What we found, and what comes next

Our first results are encouraging, but not final.

1

Check the data first

Three of the biggest "trends" were reporting breaks.

2

Cultivated land drives production

Yields rose for most crops; area decided the winners.

3

Water and heat over fertilizer

After fixed effects, irrigation and heat remain; nitrogen helps only where use is low.

NEXT STEPS

Rolling future-year tests by district → **Per-crop error reports** → **Policy extraction + testing** → **Prices only if data pass audit**

Evidence base and data sources

Full citations for the benchmark studies, data, policy context and methods.

1 **Dharmaraja, S. et al. (2020)**
Empirical analysis for crop yield forecasting in India. Agricultural Research, 9, 132–138.
doi.org/10.1007/s40003-019-00413-x

2 **Birthal, P.S. et al. (2014)**
Impact of climate change on yields of major food crops in India. Agric. Econ. Research Review, 27(2), 145–155.
doi.org/10.5958/0974-0279.2014.00019.6

3 **Jabed, M.A. & Murad, M.A.A. (2024)**
Crop yield prediction in agriculture: a review of ML and DL approaches. Heliyon, 10, e40836.
doi.org/10.1016/j.heliyon.2024.e40836

4 **Government of India, Ministry of Finance (2026)**
Economic Survey 2025–26, Chapter 6: Agriculture and Food Management.
indiabudget.gov.in/economicsurvey

5 **ICRISAT / Tata-Cornell Institute**
District-Level Database for Indian Agriculture and Allied Sectors.
icrisat.org/public/resources/data/district-level-data

6 **Gulati, A., Chatterjee, T. & Hussain, S. (2018)**
Supporting Indian farmers: price support or direct income support? ICRIER WP 357.
icrier.org/pdf/Working_Paper_357.pdf

7 **OECD (2018)**
Agricultural Policies in India. OECD Food and Agricultural Reviews.
oecd.org — Agricultural Policies in India

8 **IBEF (2026)**
Food processing industry in India.
ibef.org/industry/food-processing

9 **Angrist, J.D. & Pischke, J.-S. (2009)**
Mostly Harmless Econometrics: An Empiricist's Companion. Princeton University Press.
Princeton University Press

10 **Imbens, G.W. & Rubin, D.B. (2015)**
Causal Inference for Statistics, Social, and Biomedical Sciences. Cambridge University Press.
doi.org/10.1017/CBO9781139025751

Benchmark metrics and project results are not directly comparable unless the target, units, geography and test period match.